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Module 05. Game Theory:

Introduction:

Game Theory:

- Game theory is one way to consider the impact of the strategies of others on our strategies and outcomes.
- Game theory is the study of how optimal strategies are formulated in conflict.

Reason: to use: to determine the best strategies given a competitive business environment.

History: Started in 1944

Applications:

- ① Analysis of the market strategies of a business organization in the long run.
- ② Evaluation of the response of the consumers to a new product.
- ③ Resolving the conflict between 2 groups in a business organization.
- ④ Decision making on the techniques to increase market share.
- ⑤ Material procurement process.

Ex: Two companies A & B, sell 2 brands of flu medicine. Company A sells in 3 ways & company B sells in 4 ways. so,

8: Loss to B, Gain to A by 8.
 -2: Gain to B, Loss to A by 2.

	B1	B2	B3	B4
A1	8	-2	9	-3
A2	6	5	6	8
A3	-2	4	-9	5

Step 1: Find row minimum & see what is the max value from that.

Max value of row min values

	B1	B2	B3	B4	Row Min
	8	-2	9	-3	-3
	6	5	6	8	5
	-2	4	-9	5	-9
Column max	8	5	9	8	

↓
Minimax

Step 2: Find Maximax

from column find ~~max~~ value column max & see what is the min. value in that.

Players: The competitors/ decision in a game and

Strategies: The optimal courses of action available to a player are referred to as his strategies.

Pay-off: The outcome of playing a game is called the pay-off to the concerned player.

Pure-strategy:

Mixed-strategy:

N-person games

Payoff Matrix:

Maximin-Minimax Principle:

- The maximum value of the minimum gains is called the maximin value of the game & the corresponding strategy is called the maximin strategy. Similarly, the minimum of the maximum losses is called the minimax value of the game & the corresponding strategy is the minimax strategy.

Maximin = Minimax : Saddle point

Classification: Basis

- No. of players
- Sum of all payoffs
- No. of strategies employed.

Rectangular games: A, B have diff strategies.

- Zero-sum games means that the sum of losses for one player must equal the sum of ~~gains~~ gains for the other player.

sum of gain & loss is 0.

Two-person ~~per~~ game:
 → on 2 parties can play.

	Reds	
	Y_1	Y_2 (Newsp.)
Football X_1	3	5
Newsp. X_2	1	-2

→ Positive Number in the above table means
 X wins & Y losses.
 → Negative " " " " " "
 X losses & Y wins.

Store X	Y	Outcome
strategy	"	
X_1	Y_1	X win 3, Y losses 3
X_1	Y_2	X win 5, Y losses 5
X_2	Y_1	X win 1, Y losses 1
X_2	Y_2	X losses, Y win 2

→ $\min \max = 3$ (Two-person game)
 $\max \min = 3$

Since $\min \max = \max \min = 3$

saddle pt at 3

$V=3$ Even if played ∞ times

	Y_1	Y_2	
X_1	3	5	$\textcircled{3}$ minimax
X_2	1	-2	-2 maximin

$\textcircled{3}$ 5
~~maximin~~ minimax

at any pr of time, only 1 strategy is used by X_1 & Y_1 . known as pure strategy game.

X - 3% Gain

Y - 3% Loss

So, X is more favourable than Y
 X has 3% value, Y only 1 value

In previous table, Y is playing to minimize losses going with min loss strategy i.e. Y_1

Ex 2: Pure Strategy Games.

		Player Y strategies		min row No.
		Y_1	Y_2	
Player X's strategies	X_1	10	6	(6) minimax
	X_2	-12	2	-12 minimax
Max: Col		10	(6)	→ Maximin
N			↓ minimax	

Maximin = minimax = 2 Saddle pt
 $X \rightarrow X_1$
 $Y \rightarrow Y_2$ uses throughout the game.

$$V = 6$$

Ex: Mixed Strategy Games:

PAGE No: _____
DATE: _____

Non Row No: _____

	Y_1	Y_2	
X_1	4	2	(2) Maximin
X_2	1	10	1
Minimax (4) (1)		10	

There are 2 players X & Y both are having 2 strategies. Outcomes of the game is 4, 1, 2, 10

→

	Y_1	Y_2	
X_1	4	2	(2) → Maximin
X_2	1	10	1
(4) Minimax.		10	

As $\text{Minimax} \neq \text{Maximin}$ value
Mixed Strategy Method.

Method 1: Expected Gain Method:

Ex:

	P	$1-P$
	Y_1	Y_2
X_1 Q	4	2
X_2 $1-Q$	1	10

Step 1: Assign the probabilities

In Row $E_G \Rightarrow$

$4P + 2(1-P)$: 1st row
 $1P + 10(1-P)$: 2nd row

→ Equate this 2 eqns.

for Eqn in column

1st column = $4Q + 1(1-Q)$

2nd " = $2Q + 10(1-Q)$

$$4P + 2 - 2P = 1P + 10 - 10P$$

$$4P - 2P - 1P + 10P = 10 - 2$$

$$11P = 8, P = \frac{8}{11} \quad \checkmark$$

$$1 - P = 1 - \frac{8}{11} = \frac{3}{11}$$

Exp:
value of gain

$$V = 4P + 2(1-P)$$

$$V = 4 \times \frac{8}{11} + 2 \left(\frac{3}{11} \right) = \frac{38}{11}$$

2nd eq: also same result

$$V = 1P + 10(1-P) = \frac{38}{11} = \underline{3.46}$$

$$4Q + 1 - Q = 2Q + 10 - 10Q$$

$$4Q - Q - 2Q + 10Q = 10 - 1$$

$$Q = \frac{9}{11}$$

$$1 - Q = \frac{2}{11}$$

$$V = 4Q + 1(1-Q)$$

$$V = 4 \times \frac{9}{11} + 1 \left(\frac{2}{11} \right)$$

$$V = \frac{38}{11} \quad \checkmark = \underline{3.46}$$

$y_1 = 8/11$ will be used $8/11$ times
 $y_2 = 3/11$ " " " " $3/11$ " "
 to minimize the max. losses

$x_1 = 9/11$ " " " " $9/11$ times
 $x_2 = 2/11$ " " " " $2/11$ times
 to achieve max. profit

Ex:

	y_1	y_2	Mixed strategy game
x_1	4	2	② • minmax
x_2	0	10	0
	④	10	
	maximum		

$\text{Maximum} \neq \text{minmax} \Rightarrow \text{Mixed strategy game}$

		p	$1-p$
		y_1	y_2
x_1	Q	4	2
x_2	1-Q	0	10

Expected
Gains

$$\begin{aligned}
 &4p + 2(1-p) \\
 &0p + 10(1-p)
 \end{aligned}$$

$$\begin{aligned}
 4p + 2(1-p) &= 0p + 10(1-p) \\
 4p + 2 - 2p &= 0p + 10 - 10p \\
 4p - 2p + 10p &= 10 - 2 \\
 12p &= 8
 \end{aligned}$$

$$p = \frac{8}{12} = \frac{2}{3}$$

$$1-p = \frac{4}{12} = \frac{1}{3}$$

column:

E.G

$$\begin{aligned} 4Q + 0(1-Q) &= 1^{\text{st}} \text{ column} \\ 2Q + 10(1-Q) &= 2^{\text{nd}} \text{ "} \end{aligned}$$

$$\begin{aligned} 4Q + 0 - 0Q &= 2Q + 10 - 10Q \\ 4Q - 2Q + 10 &= 10 \end{aligned}$$

$$12Q = 10$$

$$Q = \frac{10}{12} = \frac{5}{6}$$

$$1-Q = \frac{5}{6}$$

$$V = 10(1-p)$$

$$= 10\left(\frac{1}{3}\right) = \frac{10}{3} = 3.3$$

$$V = 4Q + 0(1-Q)$$

$$= 4 \times \frac{5}{6} = \frac{20}{6} = \frac{10}{3} = 3.3$$

Final SPD even playing ∞ times

3.3 profit to x

3.3 loss to y.

$\oplus x$

$\ominus y$

Ex > P1: A/B	B1	B2	B3	B4	
A1	20	15	12	35	(12)
A2	25	14	8	10	8
A3	40	2	10	5	2
A4	-5	4	11	0	0
	40	15	(12)	35	

So, $\min \max = \max \min = 12$

saddle pt is $A \rightarrow A1$
 $B \rightarrow B3$

Find solution of game by using Saddle point.

A will be ^{using} A1 strategy

B will be using B3 strategy

12 : value of game = 12

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2nd Method

Oddments method:

③ Using the soln procedure for a mixed strategy game, solve the following game

	x_1	x_2
x_1	4	2
x_2	0	10

Step 1: Calculate oddments

Take the higher value in that row

& then subtract it from the min value

~~min~~ 1st min if we in 1st row. So, it gives oddments of other row (2nd row)